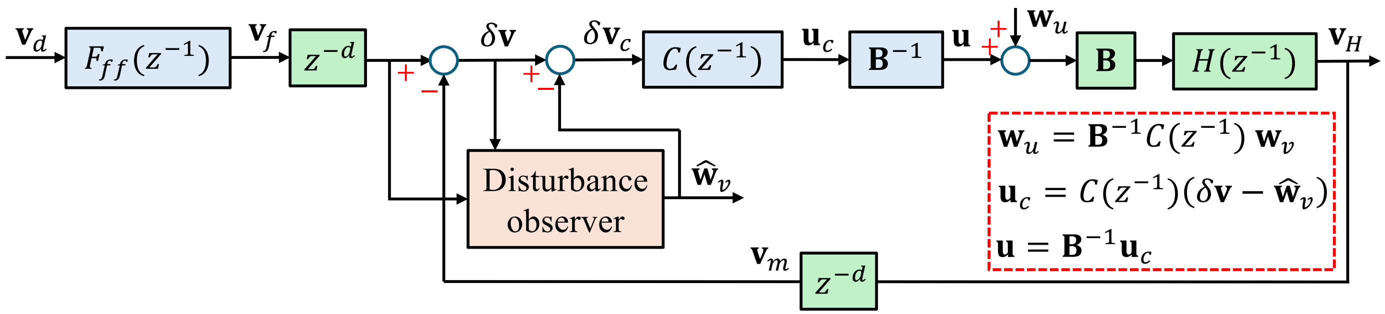

$$\frac{v_m}{u_c} = z^{-d}H(z^{-1}) = z^{-(1+d)}\frac{k_o(1+b_u z^{-1})(1+b_a z^{-1})}{(1-p_1z^{-1})(1-p_2z^{-1})(1-p_3z^{-1})} = z^{-(1+d)}\left[\frac{k_o(1+b_a z^{-1})}{1-a_1z^{-1}-a_2z^{-2}-a_3z^{-3}}\right](1+b_u z^{-1});$$

$$H_{c\ell}(z^{-1}) = \frac{z^{-d}v_H}{z^{-d}v_f} = \frac{v_H}{v_f} = \frac{z^{-d}C(z^{-1})H(z^{-1})}{1+z^{-d}C(z^{-1})H(z^{-1})} = z^{-(1+d)}\frac{k_c(1+b_u z^{-1})}{1-\lambda_c z^{-1}};\ k_c = \frac{1-\lambda_c}{1+b_u}$$

$$1 - H_{c\ell}(z^{-1}) = \frac{\delta v}{z^{-d}v_f} = 1 - z^{-(1+d)}\frac{k_c(1+b_u z^{-1})}{1-\lambda_c z^{-1}} = \frac{1-\lambda_c z^{-1}-k_c z^{-(1+d)}-b_c z^{-(2+d)}}{1-\lambda_c z^{-1}};\ b_c = b_u k_c = \frac{1-\lambda_c}{1+1/b_u}$$

$$C(z^{-1}) = \frac{1}{z^{-d}H(z^{-1})}\frac{H_{c\ell}(z^{-1})}{1-H_{c\ell}(z^{-1})} = k_u\frac{1-a_1z^{-1}-a_2z^{-2}-a_3z^{-3}}{1-b_1z^{-1}-b_{1+d}z^{-(1+d)}-b_2z^{-2}-b_{2+d}z^{-(2+d)}-b_{3+d}z^{-(3+d)}};$$

$$k_u = \frac{k_c}{k_o};\ b_1 = \lambda_c - b_a;\ b_{1+d} = k_c;\ b_2 = b_a\lambda_c;\ b_{2+d} = b_c + b_a k_c;\ b_{3+d} = b_a b_c$$

$$F_{ff}(z^{-1}) \ = k_{ff}(1-\lambda_c z^{-1})z^{-1}(1+b_u z) = k_{ff}[b_u + (1-b_u\lambda_c)z^{-1} - \lambda_c z^{-2}];$$

$$k_{ff} = \frac{1}{(1+b_u)(1-\lambda_c)};$$

$$F_{ff}(z^{-1})H_{c\ell}(z^{-1}) \ = k_f z^{-(2+d)}[1 + b_u(z+z^{-1}) + b^2]; \ k_f = \frac{1}{(1+b_u)^2}$$

$A(\theta;\ b_u) = k_f(1 + 2b_u \cdot \cos\theta \ + b_u^2)$ and $\phi(\theta) = -(2+d)\theta$

$$\mathbf{v}_f[k] = k_{ff}\{b_u \mathbf{v}_d[k] + (1-b_u\lambda_c)\mathbf{v}_d[k-1] - \lambda_c \mathbf{v}_d[k-2]\}; \quad k_{ff} = \frac{1}{(1+b_u)(1-\lambda_c)}$$

$$\delta\mathbf{v}_f[k] = \mathbf{v}_f[k-d] - \lambda_c \mathbf{v}_f[k-d-1] - k_c \mathbf{v}_f[k-2d-1] - b_c \mathbf{v}_f[k-2d-2];$$

$$k_c = \frac{1-\lambda_c}{1+b_u}; \ b_c = b_u k_c = \frac{1-\lambda_c}{1+1/b_u}$$

$$\delta\mathbf{v}[k] = \lambda_c \delta\mathbf{v}[k-1] + \delta\mathbf{v}_f[k] - k_c\left\{\mathbf{e}_{w_v}[k-d-1] + b_u \mathbf{e}_{w_v}[k-d-2]\right\}$$

Design of Control Law

Estimator ($d = 0$):

$$\delta\mathbf{v}_f[k] = \mathbf{v}_f[k] - \lambda_c\mathbf{v}_f[k-1] - k_c\mathbf{v}_f[k-1] - b_c\mathbf{v}_f[k-2]$$

$$\delta\mathbf{v}[k] = \lambda_c\delta\mathbf{v}[k-1] + \delta\mathbf{v}_f[k] - k_c\{\mathbf{e}_{w_v}[k-1] + b_u\mathbf{e}_{w_v}[k-2]\}$$

$$\delta\hat{\mathbf{v}}[k] = \lambda_c\delta\hat{\mathbf{v}}[k-1] + \delta\mathbf{v}_f[k] + \ell_1\mathbf{e}_{\delta v}[k-1]$$

$$\hat{\mathbf{w}}_1[k] = (1+\beta)\hat{\mathbf{w}}_1[k-1] - \beta\hat{\mathbf{w}}_2[k-1] + \ell_2\mathbf{e}_{\delta v}[k-1]$$

$$\hat{\mathbf{w}}_2[k] = \hat{\mathbf{w}}_1[k-1] + \ell_3\mathbf{e}_{\delta v}[k-1]$$

Uncoupled error dynamics (estimation):

$$\begin{bmatrix} e_{\delta v}[k] \\ e_{w1}[k] \\ e_{w2}[k] \end{bmatrix} = \begin{bmatrix} \lambda_c - \ell_1 & -k_c & -b_u k_c \\ -\ell_2 & 1+\beta & -\beta \\ -\ell_3 & 1 & 0 \end{bmatrix} \begin{bmatrix} e_{\delta v}[k-1] \\ e_{w1}[k-1] \\ e_{w2}[k-1] \end{bmatrix}$$

$$\begin{cases} \ell_1 = \lambda_c + (1+\beta) - 3\lambda_e \\ \ell_2 = \dfrac{b_u(\lambda_e - 1)^3 - \beta(b_u+1)(\beta^2 - 3\beta\lambda_e + \beta + 3\lambda_e^2 - 3\lambda_e + 1)}{k_c(b_u+1)(b_u+\beta)} \\ \ell_3 = \dfrac{-\beta - b_u - \beta b + 3\beta\lambda_e + 3b_u\lambda_e - \beta^2 b_u - 3b_u\lambda_e^2 - \beta^2 - \lambda_e^3 + 3\beta b_u\lambda_e}{k_c(b_u+1)(b_u+\beta)} \end{cases}$$

Estimator ($d = 1$):

$$\delta\mathbf{v}_f[k] = \mathbf{v}_f[k-1] - \lambda_c\mathbf{v}_f[k-2] - k_c\mathbf{v}_f[k-3] - b_c\mathbf{v}_f[k-4]$$

$$\begin{aligned} \delta\mathbf{v}[k] = {} & \lambda_c\delta\mathbf{v}[k-1] + \delta\mathbf{v}_f[k] \\ & - k_c\{\mathbf{e}_{w_v}[k-2] + b_u\mathbf{e}_{w_v}[k-3]\} \end{aligned}$$

$$\delta\hat{\mathbf{v}}[k] = \lambda_c\delta\hat{\mathbf{v}}[k-1] + \delta\mathbf{v}_f[k] + \ell_1\mathbf{e}_{\delta v}[k-1]$$

$$\hat{\mathbf{w}}_1[k] = (1+\beta)\hat{\mathbf{w}}_1[k-1] - \beta\hat{\mathbf{w}}_2[k-1] + \ell_2\mathbf{e}_{\delta v}[k-1]$$

$$\hat{\mathbf{w}}_2[k] = \hat{\mathbf{w}}_1[k-1] + \ell_3\mathbf{e}_{\delta v}[k-1]$$

$$\hat{\mathbf{w}}_3[k] = \hat{\mathbf{w}}_2[k-1] + \ell_4\mathbf{e}_{\delta v}[k-1]$$

Uncoupled error dynamics (estimation):

$$\begin{bmatrix} e_{\delta v}[k] \\ e_{w1}[k] \\ e_{w2}[k] \\ e_{w3}[k] \end{bmatrix} = \begin{bmatrix} \lambda_c - \ell_1 & 0 & -k_c & -b_u k_c \\ -\ell_2 & 1+\beta & -\beta & 0 \\ -\ell_3 & 1 & 0 & 0 \\ -\ell_4 & 0 & 1 & 0 \end{bmatrix} \begin{bmatrix} e_{\delta v}[k-1] \\ e_{w1}[k-1] \\ e_{w2}[k-1] \\ e_{w3}[k-1] \end{bmatrix}$$

Implementation of Control Law ($d = 0$)

Input: $\mathbf{v}_d[k]$

Measurement: $\mathbf{v}_m[k]$ Acquired from ADC

Real time computation:

1. $\mathbf{v}_f[k] = k_{ff}\{b_u\mathbf{v}_d[k] + (1 - b_u\lambda_c)\mathbf{v}_d[k-1] - \lambda_c\mathbf{v}_d[k-2]\}$
2. $\delta\mathbf{v}_f[k] = \mathbf{v}_f[k] - \lambda_c\mathbf{v}_f[k-1] - k_c\mathbf{v}_f[k-1] - b_c\mathbf{v}_f[k-2]$
3. $\delta\hat{\mathbf{v}}[k] = \lambda_c\delta\hat{\mathbf{v}}[k-1] + \delta\mathbf{v}_f[k] + \ell_1\mathbf{e}_{\delta v}[k-1]$
4. $\hat{\mathbf{w}}_1[k] = (1+\beta)\hat{\mathbf{w}}_1[k-1] - \beta\hat{\mathbf{w}}_2[k-1] + \ell_2\mathbf{e}_{\delta v}[k-1]$
5. $\hat{\mathbf{w}}_2[k] = \hat{\mathbf{w}}_1[k-1] + \ell_3\mathbf{e}_{\delta v}[k-1]$
6. $\delta\mathbf{v}[k] = \mathbf{v}_f[k] - \mathbf{v}_m[k]$
7. $\delta\mathbf{v}_c[k] = \delta\hat{\mathbf{v}}[k] - \hat{\mathbf{w}}_1[k]$
8. $\mathbf{u}_c[k] = b_1\mathbf{u}_c[k-1] + b_{1+d}\mathbf{u}_c[k-1] + b_2\mathbf{u}_c[k-2] + b_{2+d}\mathbf{u}_c[k-2] + b_{3+d}\mathbf{u}_c[k-3] + k_u\{\delta\mathbf{v}_c[k] - a_1\delta\mathbf{v}_c[k-1] - a_2\delta\mathbf{v}_c[k-2] - a_3\delta\mathbf{v}_c[k-3]\}$
9. $\mathbf{u}[k] = \mathbf{B}^{-1}\mathbf{u}_c[k]$ Sent to DAC
10. $\mathbf{e}_{\delta v}[k-1] = \delta\mathbf{v}[k] - \delta\hat{\mathbf{v}}[k]$
11. $\mathbf{v}_d[k-2] = \mathbf{v}_d[k-1]$
12. $\mathbf{v}_d[k-1] = \mathbf{v}_d[k]$
13. $\mathbf{v}_f[k-2] = \mathbf{v}_f[k-1]$
14. $\mathbf{v}_f[k-1] = \mathbf{v}_f[k]$
15. $\delta\hat{\mathbf{v}}[k-1] = \delta\hat{\mathbf{v}}[k]$
16. $\hat{\mathbf{w}}_1[k-1] = \hat{\mathbf{w}}_1[k]$
17. $\hat{\mathbf{w}}_2[k-1] = \hat{\mathbf{w}}_2[k]$
18. $\mathbf{u}_c[k-3] = \mathbf{u}_c[k-2]$
19. $\mathbf{u}_c[k-2] = \mathbf{u}_c[k-1]$
20. $\mathbf{u}_c[k-1] = \mathbf{u}_c[k]$
21. $\delta\mathbf{v}_c[k-3] = \delta\mathbf{v}_c[k-2]$
22. $\delta\mathbf{v}_c[k-2] = \delta\mathbf{v}_c[k-1]$
23. $\delta\mathbf{v}_c[k-1] = \delta\mathbf{v}_c[k]$